

BERRY–ESSEEN BOUNDS FOR DESIGN-BASED CAUSAL INFERENCE WITH POSSIBLY DIVERGING TREATMENT LEVELS AND VARYING GROUP SIZES

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Neyman (1923/1990) introduced the randomization model, which contains the notation of potential outcomes to define causal effects and a framework for large-sample inference based on the design of the experiment. However, the existing theory for this framework is far from complete, especially when the number of treatment levels diverges and the treatment group sizes vary. We provide a unified discussion of statistical inference under the randomization model with general treatment group sizes. We formulate the estimator in terms of a linear permutation statistic and use results based on Stein’s method to derive various Berry–Esseen bounds on the linear and quadratic functions of the estimator. These new Berry–Esseen bounds serve as the basis for design-based causal inference with possibly diverging treatment levels and a diverging number of causal parameters of interest. We also fill an important gap by proposing novel variance estimators for experiments with possibly many treatment levels without replications. Equipped with the newly developed results, design-based causal inference in general settings becomes more convenient with stronger theoretical guarantees.

1. Motivation: Randomization-based causal inference.

1.1. Existing results. In a seminal paper, Neyman (1923/1990) introduced the notation of potential outcomes to define causal effects. More importantly, he also proposed a framework for statistical inference of causal effects based on the design of the experiment. In particular, he considered an experiment with N units and Q treatment arms, where the number of units under treatment q equals N_q , with $\sum_{q=1}^Q N_q = N$. Corresponding to treatment level q , unit i has the potential outcome $Y_i(q)$, where $i = 1, \dots, N$ and $q = 1, \dots, Q$. Despite its simplicity, the following completely randomized experiment has been widely used in practice and has generated rich theoretical results. Definition 1 below characterizes the joint distribution of $Z = (Z_1, \dots, Z_N)$ under complete randomization, where $Z_i \in \{1, \dots, Q\}$ is the treatment indicator for unit i .

DEFINITION 1 (Complete randomization). Fix treatment group sizes $N_1, \dots, N_Q$ with $\sum_{q=1}^Q N_q = N$. The treatment vector Z is uniform over all its possible values.

Mathematically, Definition 1 implies that $\mathbb{P}(Z = z) = N_1! \cdots N_Q! / N!$ for all possible values of $z = (z_1, \dots, z_N)$ such that $\sum_{i=1}^N 1\{z_i = q\} = N_q$. Computationally, Definition 1 implies that Z is from a random permutation of N_1 1’s, $\dots$, N_Q Q ’s. Neyman (1923/1990)

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formulated complete randomization based on an urn model, which is equivalent to Definition 1. The experiment reveals one of the potential outcomes, which is the observed outcome $Y_i = Y_i(Z_i) = \sum_{q=1}^Q Y_i(q) 1\{Z_i = q\}$ for each unit i .

In Neyman (1923/1990)'s framework, all potential outcomes are fixed and only the treatment indicators are random according to Definition 1. Scheffé ((1959), Chapter 9) called it the *randomization model*. Under this model, it is conventional to call the resulting inference as *randomization inference* or *design-based inference*. It has become increasingly popular in both theory and practice (e.g., Kempthorne (1952), Copas (1973), Robins (1988), Rosenbaum (2002), Hinkelmann and Kempthorne (2007), Freedman (2008a,b), Lin (2013), Dasgupta, Pillai and Rubin (2015), Imbens and Rubin (2015), Athey and Imbens (2017), Fogarty (2018a), Guo and Basse (2023)). We focus on Neyman (1923/1990)'s framework throughout the paper.

A central goal in Neyman (1923/1990)'s framework is to use the observed data $(Z_i, Y_i)_{i=1}^N$ to make inference of causal effects defined by the potential outcomes. Define

$$\bar{Y}(q) = N^{-1} \sum_{i=1}^N Y_i(q), \quad S(q, q') = (N-1)^{-1} \sum_{i=1}^N (Y_i(q) - \bar{Y}(q))(Y_i(q') - \bar{Y}(q'))$$

as the average value of the potential outcomes under treatment q and the covariance of the potential outcomes under treatments q and q' , respectively. Define the average potential outcome vector as $\bar{Y} = (\bar{Y}(1), \dots, \bar{Y}(Q))^T \in \mathbb{R}^Q$, and define the covariance matrix of the potential outcomes as $S = (S(q, q'))_{q, q'=1, \dots, Q}$. The parameter of interest is a linear transformation of $\bar{Y}$:

$$\gamma = F^T \bar{Y}$$

for a prespecified $F \in \mathbb{R}^{Q \times H}$. We call the matrix F the coefficient matrix. In many problems, F is a contrast matrix with columns orthogonal to $(1, \dots, 1)^T$. Despite the simple form of γ , it can answer questions from a wide range of applications. For instance, Neyman (1923/1990) considered pairwise differences in means, and Dasgupta, Pillai and Rubin (2015) and Mukerjee, Dasgupta and Rubin (2018) considered linear combinations of the mean vector. Recently, Li and Ding (2017) unified the literature by studying the properties of the moment estimator for γ under complete randomization. In particular, define the sample mean and variance of the observed $Y_i(q)$'s as

$$(1) \quad \hat{Y}_q = N_q^{-1} \sum_{Z_i=q} Y_i, \quad \hat{S}(q, q) = (N_q - 1)^{-1} \sum_{Z_i=q} (Y_i - \hat{Y}_q)^2,$$

respectively. Define

$$(2) \quad \hat{Y} = (\hat{Y}_1, \dots, \hat{Y}_Q)^T \in \mathbb{R}^Q, \quad \hat{V}_{\hat{Y}} = \text{Diag}\{N_q^{-1} \hat{S}(q, q)\}_{q \in [Q]} \in \mathbb{R}^{Q \times Q}$$

as the vector of sample averages and the diagonal matrix of the sample variances across all arms, respectively. Under complete randomization, the random vector $\hat{Y}$ has mean and covariance

$$(3) \quad \mathbb{E}\{\hat{Y}\} = \bar{Y}, \quad \text{Cov}\{\hat{Y}\} = V_{\hat{Y}} = \text{Diag}\{N_q^{-1} S(q, q)\}_{q \in [Q]} - N^{-1} S,$$

and moreover, $\hat{V}_{\hat{Y}}$ is a conservative estimator for $V_{\hat{Y}}$ in the sense that $\mathbb{E}\{\hat{V}_{\hat{Y}}\} - V_{\hat{Y}} = N^{-1} S$ is positive semidefinite (see Li and Ding (2017) for a review). Therefore,

$$(4) \quad \hat{\gamma} = F^T \hat{Y}, \quad \hat{V}_{\hat{\gamma}} = F^T \hat{V}_{\hat{Y}} F$$

are an unbiased point estimator for γ and a conservative covariance estimator for $V_{\hat{\gamma}} = \text{Cov}\{\hat{\gamma}\} = F^T V_{\hat{Y}} F$, respectively. Li and Ding (2017) also used the established combinatorial or rank central limit theorems (CLTs) (Hájek (1960), Hoeffding (1951), Fraser (1956)) to prove the asymptotic normality of $\hat{\gamma}$ and the validity of the associated large-sample Wald-type inference, under certain regularity conditions.

1.2. *Open questions.* Despite the long history of [Neyman \(1923/1990\)](#)'s randomization model, the theory for randomization-based causal inference is far from complete. Technically, [Li and Ding \(2017\)](#)'s review only covered the first regime (R1) below, and even there, finer results such as Berry–Esseen bounds (BEBs) have not been rigorously established for the most general setting. For other regimes below, many basic results are still missing in the literature.

(R1) Small Q and large N_q 's. In this regime, the number of arms is small and the sample size in each arm is large. Asymptotically, as $N \rightarrow \infty$, we have that Q is a fixed integer and $N_q/N \rightarrow e_q \in (0, 1)$ for all $q = 1, \dots, Q$. [Li and Ding \(2017\)](#) showed that, under (R1) and some regularity conditions on the potential outcomes, we have

$$(5) \quad V_{\hat{\gamma}}^{-1/2}(\hat{\gamma} - \gamma) \rightsquigarrow \mathcal{N}(0, I_H), \quad N\hat{V}_{\hat{\gamma}} - N\mathbb{E}\{\hat{V}_{\hat{\gamma}}\} = o_{\mathbb{P}}(1),$$

which ensures that the large sample Wald-type inference based on the Normal approximation is conservative. [Li and Ding \(2017\)](#)'s results are asymptotic. An important theoretical question is to derive nonasymptotic results.

(R2) Large Q and large N_q 's. In this regime, each arm has adequate units for the variance estimation, but the number of arms is also large. Asymptotically, as $N \rightarrow \infty$, we have $Q \rightarrow \infty$ and $N_q \rightarrow \infty$ for all $q = 1, \dots, Q$. Consequently, the limiting values of some N_q/N 's must be 0. The point estimates and variance estimators in (4) are still well-defined in this regime. We might expect that the asymptotic results in (5) still hold because of large N_q 's. However, previous theoretical results do not cover this seemingly easy case due to the possibly diverging dimension of F .

(R3) Large Q and small N_q 's. In this regime, the number of arms is large but the sample size within each arm is small. Asymptotically, as $N \rightarrow \infty$, we have $Q \rightarrow \infty$ and $2 \leq N_q \leq \bar{n}$ for all q 's and some fixed $\bar{n} \geq 2$. This regime is well suited for many factorial experiments (see Example 1 below), in which the total number of factor combinations can be much larger than the number of replications in each combination (e.g., [Mukerjee and Wu \(2006\)](#), [Wu and Hamada \(2021\)](#)). Although the point estimate and variance estimator in (4) are still well-defined, we do not expect a simple CLT based on the joint asymptotic normality of $\hat{Y}$ due to the small N_q 's. Nevertheless, $\hat{\gamma} = F^\top \hat{Y}$, as a linear transformation of $\hat{Y}$, can still satisfy the CLT for some choice of F . This regime is a reminiscence of the so-called *proportional asymptotics* in regression analysis, and even there, statistical inference is still not satisfactory in general (e.g., [El Karoui et al. \(2013\)](#), [Lei, Bickel and El Karoui \(2018\)](#), [El Karoui and Purdom \(2018\)](#)). Technically, we need to analyze $F^\top \hat{Y}$ with the dimension of $\hat{Y}$ proportional to the sample size under the randomization model. This is a gap in the literature.

(R4) Large Q and $N_q = 1$ for all $q = 1, \dots, Q$. This regime is much harder than (R3) because the variance estimator in (4) is not even well-defined due to the lack of replications within each arm (e.g., [Espinosa, Dasgupta and Rubin \(2016\)](#)). Therefore, we need to answer two fundamental questions. First, does $F^\top \hat{Y}$ still satisfy the CLT for some F ? Second, how do we estimate the variance of $F^\top \hat{Y}$? These two questions are the basis for large-sample Wald-type inference in this regime. Neither has been covered by existing results.

(R5) Mixture of (R1)–(R4). In the most general case, it is possible that the number of treatment levels diverges and the group sizes within different treatment arms vary a lot. Theoretically, we can partition the treatment levels into different types corresponding to the four regimes above. Understanding (R5) relies on understanding (R1)–(R4). Due to the difficulties in (R1)–(R4) mentioned above, a rigorous analysis of (R5) requires a deeper understanding of the randomization model. This is another gap in the literature.

1.3. *Classification of designs based on treatment group sizes.* For descriptive convenience, we define (R1)–(R4) as *nearly uniform designs* and (R5) as *general designs*, respectively, based on the heterogeneity of the group sizes across treatment arms. Definitions 2 and 3 below make the intuition more precise.

DEFINITION 2 (Nearly uniform design). There exists a positive integer $N_0 > 0$ and absolute constants $\underline{c} \leq \bar{c}$, such that $N_q = c_q N_0$ with $\underline{c} \leq c_q \leq \bar{c}$, for all $q = 1, \dots, Q$.

Definition 2 is a finite-sample characterization. It can allow N_0 to grow with N as in (R1) and (R2); it can also allow N_0 to be fixed as in (R3) and (R4) with a growing number of treatment levels Q . It is a unified description of (R1)–(R4) where each arm contains a similar number of replications.

DEFINITION 3 (General design). Partition the treatment arms as $\{1, \dots, Q\} = Q_S \cup Q_L$ with detailed descriptions below:

- (i) Q_L contains the arms with large sample sizes. There exists a positive integer N_0 and absolute constants $\underline{c} \leq \bar{c}$, such that $N_q = c_q N_0$ with $\underline{c} \leq c_q \leq \bar{c}$, for all $q \in Q_L$.
- (ii) Q_S contains the arms with small sample sizes. There exists a fixed integer $\bar{n}$ such that $N_q \leq \bar{n}$ for all $q \in Q_S$. Further partition Q_S as $Q_S = Q_U \cup Q_R$ where
 - Q_R contains the arms with replications, that is, $2 \leq N_q \leq \bar{n}$ for all $q \in Q_R$;
 - Q_U contains the arms without replications, that is, $N_q = 1$ for all $q \in Q_U$.

For simplicity, we will use $|Q_\star|$ and $N_\star = \sum_{q \in Q_\star} N_q$ to denote the number of arms and the sample size in $Q_\star$, respectively, where $\star \in \{S, U, R, L\}$. As a special case of Definition 3, $|Q_R| = |Q_L| = 0$ corresponds to *unreplicated designs* in which each treatment level has only one observation.

Definition 3 gives a partition of the treatment levels. It is a finite-sample characterization and does not impose any restrictions on the magnitude of N_0 and $\bar{n}$. Nevertheless, it is indeed motivated by the regime in which the arms in Q_L contain many replications and the arms in Q_S contain small numbers of replications. In other words, Definition 3 is more interesting for the regime in which N_0 is much larger than $\bar{n}$. Our way of defining Q_L provides a simple way to decipher the finite-sample implications of the derived BEBs. In the main theoretical results developed later, we will impose further restrictions on N_0 and $\bar{n}$.

In practice, for most experiments, the partition is naturally dictated by the design protocol and can therefore be fixed *a priori*. For example, in the partially nested experiment discussed in Example 2 later, treated and control units are naturally categorized into large and small arms. In general, the partition should always be determined case by case, taking the specifics of the design into account.

Table 1 summarizes the important regimes and reviews the established and missing theoretical results. The overarching goal of the paper is to provide BEBs for all regimes in Table 1.

1.4. *Motivating examples.* There are many practical experimental settings that are relevant to our regimes. We will use the 2^K factorial design as a canonical example for many theoretical results throughout. We review the basic setup of the 2^K factorial design in Example 1 below (Dasgupta, Pillai and Rubin (2015), Lu (2016), Zhao and Ding (2022)).

EXAMPLE 1 (Factorial design). A 2^K factorial design has K binary factors, which generate $Q = 2^K$ treatment levels. Index the potential outcomes $Y_i(q)$'s also as $Y_i(z_1, \dots, z_K)$'s, where $q = 1, \dots, Q$ and $z_1, \dots, z_K \in \{0, 1\}$. The parameter of interest $\gamma = F^\top \bar{Y}$ may consist

TABLE 1

Theoretical results for multi-armed experiments under the randomization model. The regimes (R1)–(R4) correspond to nearly uniform designs by Definition 2, whereas the regime (R5) corresponds to general designs by Definition 3

Regime	Q	N_q	CLT, variance estimation, and BEB
(R1)	Small	Large	CLT and variance estimation; no BEB
(R2)	Large	Large	Seems similar to (R1) but not studied
(R3)	Large	Small but $N_q \geq 2$	Not studied
(R4)	Large	$N_q = 1$	Not studied; variance estimation is nontrivial
(R5)	Mixture of the above		Not studied

of a subset of the factorial effects. The contrast matrix F has orthogonal columns and entries of $\pm Q^{-1}$ (this differs from the standard contrast matrix by a factor of 2 but does not change the problem fundamentally). For example, when $K = 2$, the three factorial effects are characterized by the following contrast matrix:

$$F = \begin{matrix} & \tau_{\{1\}} & \tau_{\{2\}} & \tau_{\{1,2\}} \\ \begin{matrix} (00) \\ (01) \\ (10) \\ (11) \end{matrix} & \begin{pmatrix} -1 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & -1 \\ 1 & 1 & 1 \end{pmatrix} \end{matrix} / 4.$$

See [Dasgupta, Pillai and Rubin \(2015\)](#) for precise definitions of main effects and interactions in 2^K factorial experiments.

By definition, the factorial design can have a large number of treatment levels and varying treatment group sizes. Previous asymptotic results only covered factorial designs under (R1) with fixed K and large sample sizes for all treatment levels ([Li, Ding and Rubin \(2020\)](#), [Li and Ding \(2017\)](#), [Wu and Ding \(2021\)](#), [Zhao and Ding \(2022, 2023\)](#)). This asymptotic regime can be a poor approximation to finite-sample properties of factorial designs with even a moderate K (e.g., if $K = 10$ then $Q = 2^K > 1000$). Such regimes are getting popular in conjoint survey experiments in political science ([Hainmueller, Hopkins and Yamamoto \(2014\)](#), [Caughey, Katsumata and Yamamoto \(2019\)](#), [Zhirkov \(2022\)](#), [Shi, Wang and Ding \(2025\)](#)). Based on simulation, [Zhao and Ding \(\(2022\), Appendix D\)](#) showed that CLTs are likely to hold even with diverging K and small sample sizes for all treatment levels. Allowing for a diverging K , [Li and Ding \(\(2017\), Theorem A1\)](#) derived the CLT for a single factorial effect under the sharp null hypothesis of no treatment effects for any units whatsoever, that is, $Y_i(1) = \dots = Y_i(Q)$ for all $i = 1, \dots, N$. However, deriving general asymptotic results for the factorial design has been an open problem in the literature.

As another motivating example, we consider the following partially nested experiment with provider effects, which necessitates the study of general designs given by Definition 3.

EXAMPLE 2 (Partially nested experiment with provider effects). Let $q = 1, \dots, Q - 1$ index the treated arms in which only a limited number of units are recruited; for simplicity, assume $N_q \leq \bar{n}$ with a bounded $\bar{n}$ for $q \leq Q - 1$. Finally, let $q = Q$ index the control arm with many units so that $N_Q \rightarrow \infty$. Unlike the classical treatment-control experiments where only two arms are involved, in this case the treatment has many versions, due to the fact that different physicians are involved or different types/dosages of drugs are administered, etc. As an example, [Bauer, Sterba and Hallfors \(2008\)](#) studied an effectiveness trial of the Reconnecting Youth preventive intervention program, in which high-risk participants in the

intervention arm received the Reconnecting Youth treatment administered in groups, whereas high-risk participants assigned to the control arm were left ungrouped. Such experiments are called *partially nested experiments* because the treatment allocations are nested in small groups while the control arm is not. The effect of interest is

$$(6) \quad \gamma = \sum_{q=1}^{Q-1} f_q \bar{Y}(q) - \bar{Y}(Q),$$

where f_q 's are prespecified coefficients, for example, $f_q = (Q-1)^{-1}$ for $1 \leq q \leq Q-1$. The coefficient matrix in this example is a contrast vector: $F = (f_1, \dots, f_{Q-1}, -1)^\top$ with $\sum_{q=1}^{Q-1} f_q = 1$.

1.5. Our contributions. Section 1.2 has reviewed various designs and the associated open problems. In this paper, we will give a unified study of all the designs in Table 1. We further the literature in the following ways.

First, we formulate the inference problem under the randomization model in terms of linear permutation statistics of the form $\Gamma = (\Gamma_1, \dots, \Gamma_H)^\top$ with

$$\Gamma_h = \sum_{i=1}^N M_h(i, \pi(i)), \quad h = 1, \dots, H,$$

where $M_1, \dots, M_H$ are H deterministic $N \times N$ matrices and $\pi : [N] \rightarrow [N]$ is a random permutation on the set of integers $[N] = \{1, \dots, N\}$. This formulation is intuitive because the treatment assignment in Definition 1 follows from a random permutation of the treatment levels. In our analysis, different estimators correspond to different specifications of the matrices $M_1, \dots, M_H$, which depend on the potential outcomes. This formulation allows us to build upon the existing results in probability theory (Bolthausen (1984), Chatterjee and Meckes (2008)) to derive BEBs on the point estimator of the causal effect. In particular, our analysis emphasizes the dependence on the number of treatment levels and the dimension of the causal effects of interest. Importantly, we derive BEBs that can deal with general designs with varying group sizes.

Second, we establish a novel BEB on quadratic forms of the linear estimator under the randomization model. Importantly, this BEB allows the number of treatment levels to diverge, the sample sizes across treatment levels to vary, and the dimension of the causal effects of interest to diverge. It serves as the basis for the χ^2 approximation for large-sample Wald-type inference.

Third, we propose variance estimators for unreplicated designs and mixture designs that allow for the group size to be one in many treatment levels. To the best of our knowledge, the variance estimators are new in the literature of design-based causal inference, although they share some features with those in finely stratified survey sampling (e.g., Cochran (1977), Wolter (2007), Breidt, Opsomer and Sanchez-Borrego (2016)) and experiments (e.g., Abadie and Imbens (2008), Fogarty (2018b)). However, the theoretical analysis of the new variance estimators is much more challenging because of the dependence of the treatment indicators under the randomization model. We also study their probability limits and establish the theory for large-sample Wald-type inference.

Fourth, in the process of achieving the above three sets of results, we established some theoretical results and also discussed their connections to the existing results in the literature, which are potentially useful for other problems. For instance, we prove a novel BEB for linear permutation statistics over convex sets, building upon a recent result based on Stein's method (Fang and Röllin (2015)). Moreover, we have a thorough discussion on the connection among the established results in the literature (e.g., Molodtsova (1951), Bolthausen (1984), Chatterjee

and Meckes (2008), Wang and Li (2022)). We also obtain fine results on the sample moments under the randomization model. Due to the space limit, we relegate them to Appendices B and D in the Supplementary Material (Shi and Ding (2026)).

1.6. Notation. We use C to denote generic constants that may vary. Let $\Phi(t)$ denote the cumulative distribution function of a standard Normal distribution. For two sequences of numbers, a_N and b_N , let $a_N = O(b_N)$ denote $a_N \leq Cb_N$ for some positive constant $C > 0$, and let $a_N = o(b_N)$ denote $a_N/b_N \rightarrow 0$ as $N \rightarrow \infty$. Let 0_N and 1_N denote, respectively, vectors of all zeros and ones in $\mathbb{R}^N$. For two random variables X and X' , we use $X \lesssim X'$ or $X' \gtrsim X$ to represent that X' stochastically dominates X , that is, $\mathbb{P}\{X' \leq t\} \leq \mathbb{P}\{X \leq t\}$ for all $t \in \mathbb{R}$. For any covariance matrix V , let V^* denote the corresponding correlation matrix.

Consider a matrix $M = (M(h, l)) \in \mathbb{R}^{H \times H}$. Let $M(\cdot, l) \in \mathbb{R}^{H \times 1}$ and $M(h, \cdot) \in \mathbb{R}^{1 \times H}$ denote its l th column and h th row, respectively. Let $\varrho_k(M)$ denote its k th largest singular value. Specially, let $\varrho_{\max}(M)$ and $\varrho_{\min}(M)$ denote the largest and smallest singular values, respectively. Define its condition number as the ratio of its largest and smallest singular values: $\kappa(M) = \varrho_{\max}(M)/\varrho_{\min}(M)$. Let $\|M\|_F = (\sum_{h=1}^H \sum_{l=1}^H M(h, l)^2)^{1/2}$, $\|M\|_{\text{op}} = \{\varrho_{\max}(M^T M)\}^{1/2}$, $\|M\|_{p,r} = (\sum_{l=1}^H \|M(\cdot, l)\|_p^r)^{1/r} = \{\sum_{l=1}^H (\sum_{h=1}^H |M(h, l)|^p)^{r/p}\}^{1/r}$ ($1 \leq p < \infty$ and $1 \leq r < \infty$), $\|M\|_{\infty} = \max_{h,l \in [H]} |M(h, l)|$ be, respectively, the Frobenius norm, the operator norm, the $L_{p,r}$ norm and the vectorized ℓ_{∞} norm.

Design-based results rely on conditions on

$$M_N(q) = \max_{i \in [N]} |Y_i(q) - \bar{Y}(q)|, \quad (q = 1, \dots, Q),$$

which is the maximum absolute deviation from the mean for potential outcome $Y_i(q)$'s. Hájek (1960) used it in proving the CLT for simple random sampling, and Li and Ding (2017) used it in proving CLTs for design-based causal inference. It will also appear frequently in our presentation below.

2. BEBs for the moment estimator under completely randomized experiments. This section presents the BEBs for the moment estimator $\hat{\gamma}$ in (4) under completely randomized experiments. Section 2.1 presents general BEBs for linear projections of $\hat{\gamma}$. Section 2.2 further provides more discussion to facilitate the understanding of the established BEBs. Sections 2.3 and 2.4 then apply them to derive useful BEBs for nearly uniform designs and general designs, respectively.

2.1. BEBs on the moment estimator. To simplify the presentation, standardize $\hat{\gamma} = F^T \hat{Y}$:

$$(7) \quad \tilde{\gamma} = V_{\hat{\gamma}}^{-1/2}(\hat{\gamma} - \gamma) \quad \text{with } \mathbb{E}\{\tilde{\gamma}\} = 0 \text{ and } \text{Cov}\{\tilde{\gamma}\} = I_H.$$

The standardization (7) assumes that the covariance matrix $V_{\hat{\gamma}}$ is not singular. We assume it for convenience and it holds for most interesting cases. Nevertheless, there are special cases under which $V_{\hat{\gamma}}$ becomes degenerate. One possible reason is that the coefficient matrix F has linearly dependent columns. For such F , we can find a subset of contrast vectors to establish the BEB, and the rest is just a linear combination. Another reason for a singular $V_{\hat{\gamma}}$ is that the covariance structure of the potential outcomes might be degenerate and the contrast matrix F aligns with the null space of $V_{\hat{\gamma}}$, which are usually extreme cases that are less of interest. For estimating average treatment effect in two-arm randomized experiments, a degenerating estimator corresponds to the setting where the potential outcomes are perfectly negatively correlated (see Example 3 below), which is a rare setup in reality.

Our key results are BEBs on linear projections of $\tilde{\gamma}$. Theorem 1 below gives a general BEB for $\tilde{\gamma}$.

THEOREM 1 (BEBs for linear projections of $\tilde{\gamma}$). *Assume complete randomization.*

(i) *There exists a universal constant $C > 0$, such that for any $b \in \mathbb{R}^H$ with $\|b\|_2 = 1$, we have*

$$\sup_{t \in \mathbb{R}} |\mathbb{P}\{b^\top \tilde{\gamma} \leq t\} - \Phi(t)| \leq C \|b^\top V_{\tilde{\gamma}}^{-1/2} F^\top\|_\infty \cdot \max_{q \in [Q]} N_q^{-1} M_N(q).$$

(ii) *Further assume that there exists $\sigma_F \geq 1$ such that the covariance matrix $V_{\tilde{\gamma}}$ satisfies*

$$(8) \quad V_{\tilde{\gamma}} = F^\top V_{\hat{\gamma}} F \geq \sigma_F^{-2} F^\top \text{Diag}\{N_q^{-1} S(q, q)\} F.$$

Then there exists a universal constant $C > 0$, such that

$$(9) \quad \sup_{b \in \mathbb{R}^H, \|b\|_2=1} \sup_{t \in \mathbb{R}} |\mathbb{P}\{b^\top \tilde{\gamma} \leq t\} - \Phi(t)| \leq C \max_{i \in [N], q \in [Q]} \min\{I(i, q), II(i, q)\},$$

where

$$(10) \quad \begin{aligned} I(i, q) &= \sigma_F \left| \frac{Y_i(q) - \bar{Y}(q)}{\sqrt{N_q S(q, q)}} \right|, \\ II(i, q) &= \sigma_F \frac{\|F(q, \cdot)\|_2 \cdot N_q^{-1} |Y_i(q) - \bar{Y}(q)|}{\sqrt{\varrho_{\min}(F^\top \text{Diag}\{N_q^{-1} S(q, q)\} F)}}. \end{aligned}$$

We relegate the proof of Theorem 1 in Appendix E.1 of the Supplementary Material. To prove Theorem 1(i), we formulate $\tilde{\gamma}$ as a multivariate linear permutation statistic and apply an existing BEB by Bolthausen (1984) to obtain the BEB for $\tilde{\gamma}$. To prove Theorem 1(ii), we need to further derive upper bounds in terms of I and II in (10) from two different perspectives, which is nontrivial to the best of our knowledge.

- Term I is motivated by a careful analysis of each part of the BEB bound in Theorem 1(i) for the large arms, which summarizes an overall $N_q^{-1/2}$ rate, while the rest pieces mainly depend on the size of σ_F and the scale of the standardized potential outcomes.
- Term II is motivated by the case where there are many small arms with small N_q 's but a large Q . In this case, the inverse square root variance matrix $V_{\tilde{\gamma}}^{-1/2}$ is small due to the existence of many such small arms, and the rest pieces depend on the scale of the design matrix and potential outcomes.

Theorem 1(ii) is the key result that is applicable in a wide range of designs.

2.2. *Understanding Theorem 1.* In this subsection, we discuss Theorem 1 from several aspects.

First, we emphasize the applicability of Theorem 1 in a wide range of settings. The upper bound in Theorem 1(i) depends on the choice of b , whereas the upper bound in Theorem 1(ii) is uniform over all b . Theorem 1(i) is useful when either the infinity norm of the scaled inverse variance $\|b^\top V_{\tilde{\gamma}}^{-1/2} F^\top\|_\infty$ or the maximum term $\max_{q \in [Q]} N_q^{-1} M_N(q)$ shrinks to zero, even though the other part does not shrink to zero. Moreover, Theorem 1(ii) covers a wide range of design regimes. Technically, the upper bound in Theorem 1(ii) depends on the minimum value of two terms. It is convenient to apply these two terms to different treatment arms based on the structure of the design. We elaborate this idea by revisiting (R1) to (R5).

- For (R1) and (R2), because the N_q 's are large, we can use term I in (10) and obtain a sufficient condition for a vanishing upper bound.

- For (R3) and (R4), the N_q 's are bounded and term I in (10) has constant order. However, term II in (10) is small under mild conditions on F . For instance, in the factorial design in Example 1 above, the following algebraic facts hold:

$$(11) \quad \|F\|_\infty = Q^{-1}, \quad \|F(q, \cdot)\|_2 = Q^{-1}\sqrt{H}, \quad \varrho_{\min}(F^\top F) = Q^{-1}.$$

Combining (8) and (11), we have

$$\begin{aligned} \varrho_{\min}(F^\top V_{\hat{F}} F) &\geq \sigma_F^{-2} \varrho_{\min}(F^\top \text{Diag}\{N_q^{-1} S(q, q)\} F) \\ &\geq \sigma_F^{-2} \min_{q \in [Q]} \{N_q^{-1} S(q, q)\} \cdot \varrho_{\min}(F^\top F) \\ &= Q^{-1} \sigma_F^{-2} \min_{q \in [Q]} \{N_q^{-1} S(q, q)\}. \end{aligned}$$

Assume $\max_{q \in [Q]} M_N(q)^2 / \min_{q \in [Q]} S(q, q)$ is of constant order. Because the N_q 's are bounded, term II in (10) has order $O(\sqrt{H/Q})$, which converges to zero if $H/Q \rightarrow 0$.

- For (R5), we can partition the treatment arms based on the sizes of N_q 's to achieve a trade-off between terms I and II in (10). In particular, for general designs in Definition 3, a natural partition is $[Q] = \mathcal{Q}_L \cup \mathcal{Q}_S$. The arms in $\mathcal{Q}_L$ contain many units, so term I in (10) vanishes asymptotically. By contrast, $\mathcal{Q}_S$ contains many arms, which makes term II in (10) small under mild conditions on F .

We will provide rigorous results in the next two sections by applying Theorem 1 to obtain useful BEBs for different designs.

Second, we highlight some important theoretical insights of Theorem 1(ii) regarding a trade-off between the nonuniformity of the design and the estimand regularity. In order to translate Theorem 1 into useful asymptotic results, we need additional regularity conditions that reflect and respect the trade-off between the key quantities (N_q 's, Q and F) so that term I or term II vanish asymptotically. For those arms with large N_q 's, the scale of $F(q, \cdot)$ and Q does not impact the estimation to a great amount, because term I in (9) suggests a vanishing upper bound regardless of the choice of Q and F . In other words, we can handle more general coefficients when more replications are available. For those arms with smaller N_q 's, we have to resort to term II of (9), which requires the ratio $\|F(q, \cdot)\|_2 / (\varrho_{\min}\{F^\top V_{\hat{F}} F\})^{1/2}$ to vanish. This implies that most of the rows of the coefficient matrix (i.e., $F(q, \cdot)$'s) corresponding to the small arms should be nonzero and close in scale. We will formalize these intuitions in Sections 2.3 and 2.4 below.

Third, we add some discussion of the additional condition (8) that appeared in Theorem 1(ii). Condition (8) involves the potential outcomes and the linear coefficient jointly. In the specific designs, we can impose conditions separately on the potential outcomes and the linear contrast while keeping all our theoretical results valid. The role of Condition (8) is to help with the derivation of the upper bound in Theorem 1(ii). Condition (8) is useful for both terms I and II in (9). For term I, the parameter σ_F^2 directly appears in the expression of $I(i, q)$ in (10). For term II, it is used to further bound the denominator of $II(i, q)$ in the corollaries in Section 2.3. Condition (8) requires the covariance matrix $V_{\hat{F}}$ to be *well conditioned* in the sense that the positive definite part of $V_{\hat{F}}$ plays the dominant role. Recall that the covariance matrix formula of $V_{\hat{F}}$ has two parts: a positive definite part, $F^\top \text{Diag}\{N_q^{-1} S(q, q)\}_{q \in [Q]} F$, and a negative definite part, $-N^{-1} F^\top S F$. Condition (8), coupled with the covariance formula in (3), implies that

$$\sigma_F^{-2} F^\top \text{Diag}\{N_q^{-1} S(q, q)\}_{q \in [Q]} F \preceq V_{\hat{F}} \preceq F^\top \text{Diag}\{N_q^{-1} S(q, q)\}_{q \in [Q]} F,$$

that is, the covariance matrix $V_{\hat{F}}$ is upper and lower bounded by $F^\top \text{Diag}\{N_q^{-1} S(q, q)\}_{q \in [Q]} F$, up to constants. Condition (8) is a regularity condition that rules out counterexamples that involve extreme choices of F and S , and lead to an ill-conditioned covariance

structure. See the counterexample in Example 4 below. In general, violation of (8) occurs when the potential outcomes are highly correlated and lead to an ill-conditioned covariance S , which we need to rule out. Condition (8) holds in a wide range of practically interesting settings. Below we give some canonical examples, as well as two general sufficient conditions for (8).

EXAMPLE 3 (Two-arm randomized experiments). In treatment-control experiments, we are interested in estimating the average treatment effect $\gamma = \bar{Y}(1) - \bar{Y}(0)$, with contrast vector $F = (1, -1)^\top$. The difference-in-means estimator is $\hat{\gamma} = \hat{Y}_1 - \hat{Y}_0$. We can compute

$$V_{\hat{\gamma}} = \frac{p_0}{N_1} S(1, 1) + \frac{p_1}{N_0} S(0, 0) + \frac{2}{N} S(1, 0),$$

where $p_1 = N_1/N$ and $p_0 = N_0/N$. We can verify that (8) is equivalent to ruling out the scenario where the potential outcomes are perfectly negatively correlated (i.e., there exists a constant $c > 0$ such that $Y_i(0) = -cY_i(1)$ for all $i \in [N]$). Similar conditions also appear in existing literature, for example, Assumption 3 in Lei and Ding (2021). See Appendix E.24 for detailed justification.

EXAMPLE 4 (A counterexample in unreplicated 2^K factorial designs). We give a counterexample in unreplicated 2^K factorial designs, which have $N_q = 1$ for all $q \in [Q]$. Consider the one-dimensional contrast:

$$F = Q^{-1} (1_{2^{K-1}}^\top, -1_{2^{K-1}}^\top)^\top.$$

Let S be the following positive semidefinite matrix:

$$S = \begin{pmatrix} 1_{2^{K-1} \times 2^{K-1}} & -1_{2^{K-1} \times 2^{K-1}} \\ -1_{2^{K-1} \times 2^{K-1}} & 1_{2^{K-1} \times 2^{K-1}} \end{pmatrix}.$$

Then we can verify that $V_{\hat{\gamma}} = 0$ but $F^\top \text{Diag}\{S(q, q)\}_{q \in [Q]} F = Q^{-1} > 0$ is positive. This is a degenerate case with no uncertainty in $\hat{\gamma}$.

We conclude this section with Lemma 1 below, which gives two sufficient conditions for (8) to aid the understanding.

LEMMA 1 (Sufficient conditions for (8)). (i) Additive effects. *Condition (8) holds with $\sigma_F = 1$ if the individual causal effects are constant, that is, $F^\top (Y_i(q) - \bar{Y}(q))_{q=1}^Q = 0$ for all $i \in [N]$.* (ii) Well-conditioned correlation matrix. *Condition (8) holds with $\sigma_F = c\sigma$ if $\max_{q \in [Q]} N_q \leq (1 - c)N$ for some $0 < c < 1$ and the condition number of the correlation matrix corresponding to $V_{\hat{\gamma}}$ is upper bounded by σ^2 .*

The sufficient conditions in Lemma 1 are somewhat standard in the literature, especially under (R1). Li and Ding ((2017), Corollary 2) gives a CLT under the assumption of constant individual causal effects, which is a special case of Lemma 1(i). Lemma 1(i) ensures that under the sharp null hypothesis with $Y_i(q) = Y_i$ for $q \in [Q]$, Condition (8) holds for all of (R1)–(R5). Lemma 1(ii) also generalizes the classical results under (R1). Li and Ding ((2017), Theorem 5) proves a CLT under the assumption that S has a finite limiting value. When the limit is positive definite, $V_{\hat{\gamma}}$ also converges to a positive definite matrix, which becomes a special case of Lemma 1(ii). In general, Lemma 1(ii) can cover many other interesting scenarios. For example, with Lemma 1(ii) we can verify that if the potential outcomes from different treatment arms are uncorrelated with $S(q, q') = 0$ for all $q \neq q'$ and the sample sizes from different arms satisfy $\max_{q \in [Q]} N_q \leq (1 - c)N$ for some $0 < c < 1$, then Condition (8) holds with $\sigma_F^{-2} = c$. The general forms of conditions in (8) and Lemma 1 are useful for all of (R1)–(R5).

2.3. *A BEB with a proper coefficient matrix in nearly uniform designs.* In (R1), with a fixed Q and large N_q 's, it is intuitive to have CLTs for linear transformations of $\hat{Y}$ because $\hat{Y}$ itself has a CLT. In other regimes, for instance, (R4), CLTs for linear transformations of $\hat{Y}$ are less intuitive. Consider a diverging Q and bounded N_q 's. If $F = (1, -1, \dots, 0)^\top \in \mathbb{R}^Q$, then the CLT for $F^\top \hat{Y} = \hat{Y}_1 - \hat{Y}_2$ does not hold due to the bounded sample size in treatment arms 1 and 2. As another toy example, if

$$F = \begin{pmatrix} 1 & -1 & 0 & \cdots & 0 \\ 1 & -1 & 0 & \cdots & 0 \end{pmatrix}^\top \in \mathbb{R}^{Q \times 2},$$

then $F^\top \hat{Y}$ has degenerate covariance structure and Theorem 1 cannot be directly applied. Therefore, CLTs should be established for proper coefficient matrices. Corollary 1 below gives a BEB for proper coefficient matrices. We first introduce Condition 1 below on the coefficient matrix F .

CONDITION 1 (Proper coefficient matrix in nearly uniform designs). The coefficient matrix F satisfies $\|F\|_\infty \leq cQ^{-1}$ and $\varrho_{\min}\{F^\top F\} \geq c'Q^{-1}$ for some constants $c, c' > 0$.

Condition 1 depends on the scale of F , although the BEB should not depend on the scale of F due to the standardization of $\hat{\gamma}$. We present the above form of Condition 1 to facilitate the discussion of the factorial design in Example 1, in which the scale of F is motivated by scientific questions of interest. When Q is fixed, Condition 1 holds if F has full column rank. So in (R1), Condition 1 does not impose any additional assumptions beyond the standard ones. When Q diverges, Condition 1 rules out sparse F that only results in a linear combination of $\hat{Y}$ over a small number of treatment arms. Also, the minimum eigenvalue condition in Condition 1 ensures the nondegenerate covariance structure of the estimator $\hat{\gamma}$.

We then give Corollary 1 below.

COROLLARY 1 (BEB for nearly uniform designs). Assume complete randomization that satisfies Definition 2 and Condition 1. Also assume (8). There exists a universal constant $C > 0$, such that

$$(12) \quad \sup_{b \in \mathbb{R}^H, \|b\|_2=1} \sup_{t \in \mathbb{R}} |\mathbb{P}\{b^\top \tilde{\gamma} \leq t\} - \Phi(t)| \leq C\sigma_F \frac{\max_{q \in [Q]} M_N(q)}{\{\min_{q \in [Q]} S(q, q)\}^{1/2}} \sqrt{\frac{H}{N}}.$$

We relegate the proof of Corollary 1 to Appendix E.4 in the Supplementary Material. Technically, we derive the upper bound in Corollary 1 based on the upper bound from term II in (10) in Theorem 1(ii). We first upper bound the numerator of term II using the condition that $\|F\|_\infty$ is upper bounded by cQ^{-1} . We then lower bound the denominator using (8) and the fact that $\varrho_{\min}\{F^\top F\} \geq c'Q^{-1}$ by Condition 1.

We make several further comments on Corollary 1. First, Theorem 1(ii) is uniform over b . Therefore, the upper bound in Corollary 1 preserves the uniformity and does not depend on b . Second, the upper bound in (12) reveals the interplay of several quantities: the number of parameters H , the number of units N , the scale of the potential outcomes $M_N(q)$, the minimum second moments $\min_{q \in [Q]} S(q, q)$ as well as the structure of F . Third, the upper bound in (12) decreases at the rate of $(H/N)^{1/2}$. Under regimes (R1)–(R4), QN_0 and N have the same order as $N \rightarrow \infty$. To ensure convergence in distribution in (12), we only require H to be small compared with N , or, equivalently, H to be small compared with QN_0 . Importantly, there is no further restriction on Q or N_0 , as long as $H/(QN_0)$ converges to 0. Therefore, Corollary 1 is applicable for regimes (R1)–(R4). Fourth, the denominator of (12) depends on $\min_{q \in [Q]} S(q, q)$, which is useful when the variances of the potential

outcomes are lower bounded. For ease of presentation, we do not discuss more complicated cases such as some $S(q, q)$'s are small. We can slightly modify the proof of Corollary 1 to cover scenarios where some $S(q, q)$'s are close or equal to zero. See Remark S2 in Appendix E.4 of the Supplementary Material for more detailed discussion.

Example 5 below gives a more concrete application of Corollary 1 in the nearly uniform factorial design.

EXAMPLE 5 (Nearly uniform factorial design). Recall Example 1 and assume it satisfies Definition 2. Let $F \in \mathbb{R}^{Q \times H}$ with $H = K + K(K - 1)/2 = K(K + 1)/2$ be the coefficient matrix for all main effects and two-way interactions. The coefficient matrix F satisfies $\|F\|_\infty = Q^{-1}$ and $\varrho_{\min}\{F^\top F\} = Q^{-1}$, hence Condition 1 holds with $c = c' = 1$. Assume (8) and recall (11). Corollary 1 implies

$$(13) \quad \sup_{b \in \mathbb{R}^H, \|b\|_2=1} \sup_{t \in \mathbb{R}} |\mathbb{P}\{b^\top \tilde{\gamma} \leq t\} - \Phi(t)| \leq C \sigma_F \frac{\max_{q \in [Q]} M_N(q)}{\{\min_{q \in [Q]} S(q, q)\}^{1/2}} \sqrt{\frac{K^2}{N}}.$$

From (13), we can obtain a sufficient condition for the upper bound to converge to 0, which implies a CLT of $\tilde{\gamma}$.

2.4. A BEB for general designs. Now consider general designs in Definition 3. Analogous to our discussion in nearly uniform designs, we impose a condition on the coefficient matrix. Partition the coefficient matrix F into F_S and F_L , and further partition F_S into F_U and F_R . So, we have

$$(14) \quad F = \begin{pmatrix} F_S \\ F_L \end{pmatrix} \quad \text{where } F_S = \begin{pmatrix} F_U \\ F_R \end{pmatrix}.$$

Here, F_S, F_L, F_U, F_R are submatrices of F corresponding to the columns indexed by treatment arms in $\mathcal{Q}_S, \mathcal{Q}_L, \mathcal{Q}_U, \mathcal{Q}_R$, respectively.

CONDITION 2 (Proper coefficient matrices in general designs). The submatrix F_S of the coefficient matrix F satisfies $\|F_S\|_\infty \leq c|\mathcal{Q}_S|^{-1}$ and $\varrho_{\min}\{F_S^\top F_S\} \geq c'|\mathcal{Q}_S|^{-1}$ for some constants $c, c' > 0$.

Condition 2 is similar to Condition 1. However, Condition 2 imposes restrictions on the submatrix F_S , whereas Condition 1 imposes restrictions on the whole matrix F . Importantly, Condition 2 does not impose the restrictions on F_L , which corresponds to the treatment arms with enough replications.

We can apply Theorem 1 to establish the following BEB for general designs.

COROLLARY 2 (BEB for general designs). Assume complete randomization that satisfies Definition 3 and Condition 2. Also assume (8). There exists a universal constant $C > 0$, such that

$$(15) \quad \sup_{b \in \mathbb{R}^H, \|b\|_2=1} \sup_{t \in \mathbb{R}} |\mathbb{P}\{b^\top \tilde{\gamma} \leq t\} - \Phi(t)| \leq C \sigma_F \max \left\{ \max_{q \in \mathcal{Q}_L} \frac{M_N(q)}{\sqrt{N_q S(q, q)}}, \frac{\max_{q \in \mathcal{Q}_S} M_N(q)}{\{\min_{q \in \mathcal{Q}_S} S(q, q)\}^{1/2}} \cdot \sqrt{\frac{H}{N_S}} \right\}.$$

We relegate the proof of Corollary 2 to Appendix E.5 of the Supplementary Material. To prove Corollary 2, we apply Theorem 1 in several key steps. We first partition $\mathcal{Q}$ into $\mathcal{Q}_S \cup \mathcal{Q}_L$

based on the size of the treatment arms. With the key bound in (9), we then apply term I to $\mathcal{Q}_L$ and term II to $\mathcal{Q}_S$ with some further simplifications of the denominator of term II.

The upper bound (15) is uniform over all b . It depends on the sizes of the treatment arms in a subtle way. On the one hand, for $q \in \mathcal{Q}_L$, the N_q 's are large, so the first part of (15) converges to zero if the following “local” condition holds for all $q \in \mathcal{Q}_L$:

$$\frac{M_N(q)^2}{S(q, q)} = o(N_q).$$

On the other hand, for $q \in \mathcal{Q}_S$, the N_q 's are small, but the second part of (15) still converges to zero if the following “global” condition holds:

$$\frac{\max_{q \in \mathcal{Q}_S} M_N(q)^2}{\min_{q \in \mathcal{Q}_S} S(q, q)} = o\left(\frac{H}{N_S}\right).$$

We first apply Corollary 2 to the general factorial design.

EXAMPLE 6 (An example of general factorial designs). Recall Example 1. Assume the baseline arm $q = 1$ contains a large number of units possibly due to lower cost while the other arms have $N_q \leq \bar{n}$ for some fixed $\bar{n}$. This gives a general design by Definition 3 with $\mathcal{Q}_L = \{1\}$ and $\mathcal{Q}_S = \{2, \dots, Q\}$. Let $F \in \mathbb{R}^{Q \times H}$ with $H = K(K+1)/2$ be the contrast matrix for all main effects as well as two-way interactions. Assume (8) and recall (11). Condition 2 holds naturally for large Q , because we have $\|F_S\|_\infty = Q^{-1} \leq |\mathcal{Q}_S|^{-1}$ in factorial designs and the eigenvalue of $F_S^\top F_S$ can be lower bounded as follows:

$$\begin{aligned} \varrho_{\min}\{F_S^\top F_S\} &\geq \varrho_{\min}\{F^\top F\} - \varrho_{\max}\{F_L^\top F_L\} \\ &= Q^{-1} - Q^{-2}H = Q^{-1}(1 - HQ^{-1}) = O(Q^{-1}) = O(|\mathcal{Q}_S|^{-1}). \end{aligned}$$

Applying Corollary 2, we have

$$\begin{aligned} (16) \quad &\sup_{b \in \mathbb{R}^H, \|b\|_2=1} \sup_{t \in \mathbb{R}} |\mathbb{P}\{b^\top \tilde{\gamma} \leq t\} - \Phi(t)| \\ &\leq C\sigma_F \max\left\{ \frac{M_N(1)}{\sqrt{N_1 S(1, 1)}}, \frac{\max_{q \geq 2} M_N(q)}{\{\min_{q \geq 2} S(q, q)\}^{1/2}} \sqrt{\frac{K^2}{N_S}} \right\}. \end{aligned}$$

From (16), if $K \rightarrow \infty$, $N_1 \rightarrow \infty$, and

$$\frac{M_N(1)}{\sqrt{S(1, 1)}} = o(N_1^{1/2}), \quad \frac{\max_{q \geq 2} M_N(q)}{\{\min_{q \geq 2} S(q, q)\}^{1/2}} = o(N_S^{1/2}/K),$$

then the upper bound in (16) vanishes asymptotically.

We then apply Corollary 2 to the partially nested experiment with provider effects in Example 2 above.

EXAMPLE 7 (Revisit Example 2). Recall Example 2 with the contrast vector:

$$F = \underbrace{(f_1, \dots, f_{Q-1})}_{F_S^\top} \underbrace{(-1)}_{F_L^\top}.$$

Assume that there exists some $c, c' > 0$ such that

$$(17) \quad \|F_S\|_\infty = \max_{q \leq Q-1} |f_q| \leq c(Q-1)^{-1}, \quad \varrho_{\min}\{F_S^\top F_S\} = \sum_{q=1}^{Q-1} f_q^2 \geq c'(Q-1)^{-1},$$

which holds for the special case with $f_q = (Q - 1)^{-1}$. For general f_q 's, Condition (17) guarantees Condition 2. Intuitively, Condition (17) requires a dense number of c_q 's to be of the order $O((Q - 1)^{-1})$ so that the target parameter (6) is the contrast between the means of a weighted average of the treated arms and the control arm. Meanwhile, Condition (8) holds under many settings (see, e.g., the sufficient conditions in Lemma 1 above). The point estimator $\hat{\gamma} = \sum_{q=1}^{Q-1} f_q \hat{Y}_q - \hat{Y}_Q$ satisfies

$$(18) \quad \sup_{t \in \mathbb{R}} \left| \mathbb{P} \left\{ \frac{\hat{\gamma} - \gamma}{\sqrt{\text{Var}\{\hat{\gamma}\}}} \leq t \right\} - \Phi(t) \right| \leq C \sigma_F \max \left\{ \frac{M_N(Q)}{\sqrt{N_Q S(Q, Q)}}, \frac{\max_{q \leq Q-1} M_N(q)}{\{\min_{q \leq Q-1} S(q, q)\}^{1/2}} \sqrt{\frac{1}{N_S}} \right\}.$$

As $N_Q \rightarrow \infty$ and $N_S \rightarrow \infty$, the upper bound in (18) converges to zero if

$$\frac{M_N(Q)}{\sqrt{S(Q, Q)}} = o(N_Q^{1/2}), \quad \frac{\max_{q \leq Q-1} M_N(q)}{\{\min_{q \leq Q-1} S(q, q)\}^{1/2}} = o(N_S^{1/2}).$$

3. Design-based causal inference. Now we turn to the central task of design-based causal inference under complete randomization. We focus on the large-sample Wald-type inference based on the quadratic form

$$\hat{T} = (\hat{\gamma} - \gamma)^\top \hat{V}_{\hat{\gamma}}^{-1} (\hat{\gamma} - \gamma),$$

recalling the point estimator $\hat{\gamma}$ and the variance estimator $\hat{V}_{\hat{\gamma}}$ in (4). In (R1) with fixed (Q, H) and large N_q 's, the standard asymptotic argument suggests that we can use $q_{H, \alpha}$, the upper α -quantile of χ_H^2 , as the critical value for the quadratic form. For simplicity, we say that the corresponding confidence set is asymptotically valid if $\lim_{N \rightarrow \infty} \mathbb{P}\{\hat{T} \leq q_{H, \alpha}\} \geq 1 - \alpha$.

The rigorous theoretical justification for the above Wald-type inference procedure typically follows from two steps:

(Step 1) First, analyze the asymptotic distribution of the corresponding quadratic form with the true covariance matrix

$$(19) \quad T = (\hat{\gamma} - \gamma)^\top V_{\hat{\gamma}}^{-1} (\hat{\gamma} - \gamma).$$

(Step 2) Second, construct a consistent or conservative estimator $\hat{V}_{\hat{\gamma}}$ for the true covariance matrix $V_{\hat{\gamma}}$.

Under regime (R1), both Steps 1 and 2 have rigorous theoretical justification ensured by (5). Beyond (R1), it is challenging to derive the asymptotic distribution of the quadratic form in Step 1 especially when H , and thus the degrees of freedom of T diverge. To achieve the requirement in Step 1, we use results based on Stein's method to derive BEBs on quadratic forms of linear permutation statistics. To avoid excessive notation, we present the results that are most relevant to our inference problem in the main paper and relegate more general yet more complicated results to Appendices B and D in the Supplementary Material. Moreover, the sample variances $\hat{S}(q, q)$'s, and thus the variance estimator $\hat{V}_{\hat{\gamma}}$ in (4) are not even well-defined when some treatment arms do not have replications of the outcome. Without replications in all arms, we must find an alternative form of $\hat{V}_{\hat{\gamma}}$ to estimate $V_{\hat{\gamma}}$. This is a salient problem for (R4) and (R5). Finally, in all regimes (R1)–(R5), we need to study the properties of $\hat{V}_{\hat{\gamma}}$ to achieve the requirement in Step 2.

Due to the different levels of technical complexities, we divide this section into three subsections. Section 3.1 discusses nearly uniform designs with replications in all arms. Section 3.2 discusses unreplicated designs. Section 3.3 discusses the general designs. In every

subsection, we first present a BEB on the quadratic form in Step 1, then present the properties of the covariance estimator $\widehat{V}_{\widehat{\gamma}}$, and finally present the formal result to justify the Wald-type inference.

To facilitate the discussion, we introduce the following notation:

$$(20) \quad T_0 = \xi_H^\top \xi_H \quad \text{where } \xi_H \sim \mathcal{N}(0, I_H)$$

for a χ_H^2 random variable with possibly diverging degrees of freedom. The T_0 in (20) has mean H and variance $2H$. We will show that asymptotically with large N , the distribution of T is approximately equal to T_0 , whereas the distribution of $\widehat{T}$ is stochastically dominated by that of T_0 due to the conservativeness of the variance estimation.

We introduce the following moment condition on the potential outcomes for our theoretical analysis below.

CONDITION 3 (Bounded fourth moment of the potential outcomes). There exists an absolute constant $\Delta > 0$ such that $\max_{q \in [Q]} N^{-1} \sum_{i=1}^N \{Y_i(q) - \bar{Y}(q)\}^4 \leq \Delta^4$.

As a technical comment, we can allow Δ to grow in theory, but to keep the presentation more elegant, we assume Δ to be a constant in Condition 3. Moreover, it is possible to replace the fourth moment condition by a $\max_{q \in [Q]} N^{-1} \max_{i=1, \dots, N} |Y_i(q) - \bar{Y}(q)|^2 = o(1)$ as in [Li and Ding \(2017\)](#), when we focus on inference on a fixed set of contrasts (i.e., a fixed H). More general results are in the Supplementary Material (see Appendices E.6, E.9 and E.12 in the Supplementary Material).

3.1. Nearly uniform design with replications in all arms. In this subsection, we study the Wald-type inference for nearly uniform designs given by Definition 2. First, we present a BEB for T in (19) in Theorem 2 below.

THEOREM 2 (BEB for the quadratic form T for nearly uniform designs with replications). Assume complete randomization that satisfies Definition 2 and Condition 1. Further assume (8). There exists a universal constant $C > 0$, such that

$$(21) \quad \sup_{t \in \mathbb{R}} |\mathbb{P}(T \leq t) - \mathbb{P}(T_0 \leq t)| \leq \frac{C \max_{q \in [Q]} M_N(q)^3}{\{\min_{q \in [Q]} S(q, q)\}^{3/2}} \cdot \frac{H^{19/4}}{N^{1/2}}.$$

We relegate the proof of Theorem 2 to Appendix B.3 in the Supplementary Material. To prove Theorem 2, we first establish a general BEB over convex sets for multivariate linear permutation statistics based on [Fang and Röllin \(2015\)](#). This involves constructing an “exchangeable pair” and carrying out delicate moment calculations under complete randomization for applying [Fang and Röllin \(2015\)](#) based on Stein’s method. Theorem S2 in Appendix B.3 in the Supplementary Material presents this general BEB, which is of independent interest beyond our setting. We then apply Theorem S2 in the Supplementary Material to derive the BEB for the quadratic form T in (21).

Theorem 2 bounds the difference between the distribution of T and T_0 with possibly diverging H . Its upper bound is more useful when $H^{19/2}/N \rightarrow 0$, which restricts the number of parameters of interest. The condition $H^{19/2}/N \rightarrow 0$ holds naturally in the factorial design in Example 1 under regime (R4) if only the main effects and two-way interactions are of interest, which gives $H = O(K^2) = O((\log N)^2)$.

Second, we discuss variance estimation. Recall $\widehat{S}(q, q)$ and $\widehat{V}_{\widehat{\gamma}}$ be defined as in (1) and (2). Consider the point estimator $\widehat{\gamma}$ and covariance estimator $\widehat{V}_{\widehat{\gamma}}$ in (4). We have Theorem 3 below.

THEOREM 3 (Variance estimation in nearly uniform designs). *Consider designs that satisfy Definition 2 with $\min_{q \in [Q]} N_q \geq 2$. Assume Condition 3.*

- (i) $\mathbb{E}\{\widehat{V}_{\widehat{\gamma}}\} \succeq V_{\widehat{\gamma}}$.
- (ii) $\|\widehat{V}_{\widehat{\gamma}} - \mathbb{E}\{\widehat{V}_{\widehat{\gamma}}\}\|_{\infty}^2 = O_{\mathbb{P}}(\|F\|_{\infty}^4 Q^4 N^{-3} H^2)$.
- (iii) $\|\widehat{V}_{\widehat{\gamma}} - \mathbb{E}\{\widehat{V}_{\widehat{\gamma}}\}\|_{\text{op}}^2 = O_{\mathbb{P}}(\|F\|_{\infty}^4 Q^4 N^{-3} H^4)$.

We relegate the proof to Appendix E.7 in the Supplementary Material. Theorem 3(i) reveals that the covariance estimator $\widehat{V}_{\widehat{\gamma}}$ is conservative, which is well known in design-based causal inference (Neyman (1923/1990), Imbens and Rubin (2015), Li and Ding (2017)). We prove stronger results than Theorem 3(ii) and (iii) by establishing finite-sample tail bounds on $\widehat{V}_{\widehat{\gamma}}$ based on Chebyshev's inequality and detailed calculations of the moments under complete randomization.

Theorem 3(ii) and (iii) are novel results on the stochastic orders of the estimation error of the covariance estimator in L_{∞} norm and operator norm, respectively. In Example 1 of the factorial design with $\|F\|_{\infty} = O(Q^{-1})$, Theorem 3 simplifies to

$$N \|\widehat{V}_{\widehat{\gamma}} - \mathbb{E}\{\widehat{V}_{\widehat{\gamma}}\}\|_{\infty} = O_{\mathbb{P}}(H/N^{1/2}), \quad N \|\widehat{V}_{\widehat{\gamma}} - \mathbb{E}\{\widehat{V}_{\widehat{\gamma}}\}\|_{\text{op}} = O_{\mathbb{P}}(H^2/N^{1/2}).$$

The estimation error shrinks to zero if only the main effects and two-way interactions are of interest. The results in Theorem 3 suffice for inference, and we relegate the finer probability tail bound for $\widehat{V}_{\widehat{\gamma}}$ to Appendix E.6 in the Supplementary Material.

Third, we present formal results on inference. To simplify the presentation, we impose Condition 4 below.

CONDITION 4. (i) There exists a universal constant $\nu > 0$ that does not depend on N and Q such that $\max_{q \in [Q]} M_N(q) \leq \nu$. (ii) There exists a universal constant $\underline{S} > 0$ that does not depend on N and Q such that $\min_{q \in [Q]} S(q, q) \geq \underline{S}$.

We present Condition 4 to simplify the presentation of the theory in the main paper. More generally, we can relax Condition 4(i) on the universal upper bound on $M_N(q)$ and Condition 4(ii) on the universal lower bound on $S(q, q)$ by imposing conditions on the tail behavior of the potential outcomes. See Appendix E.23 in the Supplementary Material for more discussions.

Theorem 4 below justifies the Wald-type inference under the nearly uniform design with replications, where H can be either fixed or diverging.

THEOREM 4 (Validity of Wald-type inference under nearly uniform designs with replications). *Consider the nearly uniform design given by Definition 2 that satisfies $\min_{q \in [Q]} N_q \geq 2$ and Condition 1. Also assume (8), Conditions 3 and 4. Let $N \rightarrow \infty$. If $H^{19/2}/N \rightarrow 0$, then the Wald-type confidence set $\{\bar{\gamma} : (\widehat{\gamma} - \bar{\gamma})^{\top} \widehat{V}_{\widehat{\gamma}}^{-1} (\widehat{\gamma} - \bar{\gamma}) \leq q_{H,\alpha}\}$ for γ is asymptotically valid.*

We relegate the proof of Theorem 4 to Appendix E.7 in the Supplementary Material. To prove Theorem 4, we establish the limiting distribution of $\widehat{T}$ under both regimes with a fixed H and a diverging H . Theorem S5 in the Supplementary Material summarizes the precise results on the limiting distributions of $\widehat{T}$, which is of independent interest. The proof of Theorem S5 involves translating the finite sample bounds in Corollary 1 and Theorem 2 into asymptotic results. With a fixed H , Theorem S5 shows that $\widehat{T} \rightsquigarrow \mathcal{L}$ for some distribution $\mathcal{L}$ that is stochastically dominated by χ_H^2 . With a diverging H , Theorem S5 shows that the standardized $\widehat{T}$ converges to the standard normal distribution.

Theorem 4 extends the known result for (R1) with a fixed H and provides a novel result that allows for diverging Q and H . It relies crucially on the BEB on the quadratic form T in Theorem 2 and the stochastic properties of $\widehat{V}_{\widehat{\gamma}}$ in Theorem 3.

3.2. Unreplicated design. In this subsection, we study inference for unreplicated designs with $N_q = 1$ for $q = 1, \dots, Q$. The BEB on T is identical to that in Theorem 2. We give the formal result in Theorem 5 below for completeness.

THEOREM 5 (BEB for the quadratic form in unreplicated designs). *Assume complete randomization that satisfies $N_q = 1$ for $q = 1, \dots, Q$ and Condition 1. Also assume (8). The BEB (21) holds.*

However, covariance estimation without replications is a fundamentally challenging problem that is not unique to the design-based framework, as reviewed in Section 1. The commonly-used covariance estimator $\hat{V}_{\hat{\gamma}}$ in (4) is not well-defined. We must construct a new estimator. Related variance estimation problems appeared in stratified survey sampling and stratified randomized experiments. For stratified survey sampling, Cochran ((1977), Section 5A.12) proposed a variance estimator with one unit per stratum by grouping strata into pairs. Wolter ((2007), Section 2.5) discussed the idea of collapsing strata into groups, with a more general extension to more than two strata per group. For stratified experiments, Fogarty (2018b) discussed variance estimators based on grouping strata or incorporating covariate information. However, those theoretical analyses cannot easily generalize to a permutation distribution where observations are correlated. Many works (e.g., Hansen, Hurwitz and Madow (1953a,b), Breidt, Opsomer and Sanchez-Borrego (2016), Abadie and Imbens (2008)) also discussed using predictors to assist variance estimation, which is relevant to our problem but is beyond the scope of our paper.

Below we first consider a naive variance estimator, which is intuitive and easy for implementation. However, we will show that it is almost always strictly conservative for the true variance. As a remedy, we also propose a grouping strategy that is similar to grouping strata to form conservative variance estimators in finely stratified survey sampling and experiments.

We use slightly simplified notation for unreplicated designs, where the observed allocation Z_i and the arm q have a one-to-one correspondence. We can denote the single observed outcome in arm q by Y_q . The point estimator still has the form $\hat{\gamma} = F^\top \hat{Y}$ where $\hat{Y} = (Y_1, \dots, Y_Q)^\top$ is simply the observed outcome vector. Without replications, we cannot calculate $\hat{S}(q, q)$ based on only the single observation within arm $q \in \mathcal{Q}_U = \{1, \dots, Q\}$.

3.2.1. First strategy for variance estimation. A strategy for constructing a variance estimator in unreplicated designs is based on the fact that $\hat{\gamma}$ is the average of the random vectors:

$$\hat{\gamma} = F^\top \hat{Y} = \sum_{q \in [Q]} F(q, \cdot)^\top Y_q = Q^{-1} \sum_{q \in [Q]} Q F(q, \cdot)^\top Y_q,$$

which motivates us to construct the variance estimator:

$$(22) \quad \hat{V}_{\hat{\gamma}} = \mu_Q \sum_{q \in [Q]} (Q F(q, \cdot)^\top Y_q - \hat{\gamma})(Q F(q, \cdot)^\top Y_q - \hat{\gamma}^\top),$$

where $\mu_Q = \{Q(Q-2)\}^{-1}$ is a correction factor, which is motivated by moments calculation by the proof of Theorem 6 below.

THEOREM 6 (First strategy for variance estimation for unreplicated designs). *Consider designs that satisfy Definition 3 with $|\mathcal{Q}_R| = |\mathcal{Q}_L| = 0$ and the covariance estimator in (22).*

(i) We have

$$\begin{aligned}\mathbb{E}\{\widehat{V}_{\widehat{\gamma}}\} &= V_{\widehat{\gamma}} + \frac{Q-1}{Q(Q-2)} F^\top S F \\ &\quad + \frac{1}{Q(Q-2)} \sum_{q \in [Q]} (\gamma - QF(q, \cdot)^\top \bar{Y}(q)) (\gamma^\top - QF(q, \cdot) \bar{Y}(q)).\end{aligned}$$

Therefore, $\mathbb{E}\{\widehat{V}_{\widehat{\gamma}}\} \geq V_{\widehat{\gamma}}$.

(ii) Assume Conditions 3 and 4. We have $\|\widehat{V}_{\widehat{\gamma}} - \mathbb{E}\{\widehat{V}_{\widehat{\gamma}}\}\|_\infty^2 = O_{\mathbb{P}}(\|F\|_\infty^4 QH^2)$.

(iii) Assume Conditions 3 and 4. We have $\|\widehat{V}_{\widehat{\gamma}} - \mathbb{E}\{\widehat{V}_{\widehat{\gamma}}\}\|_{\text{op}}^2 = O_{\mathbb{P}}(\|F\|_\infty^4 QH^4)$.

Theorem 6 is established by moment calculation and the use of Chebyshev's inequality (see Appendix E.22 in the Supplementary Material). Theorem 6(i) suggests that the variance estimator (22) is conservative. The bias vanishes if and only if both of the following conditions hold:

$$\begin{aligned}F^\top (Y_i(q))_{q \in [Q]} &= \gamma \quad \text{for all } i \in [N], \\ F(q, \cdot)^\top \bar{Y}(q) &= Q^{-1} \gamma, \quad \text{for all } q \in [Q].\end{aligned}$$

These are very stringent conditions and impose restrictive conditions on the coefficient matrix F and the potential outcomes. Therefore, Theorem 6 suggests a trade-off for the use of the variance estimator (22): it is easy for implementation but almost always strictly conservative.

3.2.2. Second strategy for variance estimation. Due to the above limitation of the variance estimator (22), we consider a new strategy based on grouping the outcomes. With a little abuse of notation, we still consider the covariance estimator of the form:

$$\widehat{V}_{\widehat{\gamma}} = F^\top \widehat{V}_{\widehat{\gamma}} F,$$

where $\widehat{V}_{\widehat{\gamma}}$ is a $Q \times Q$ diagonal matrix. The key is to construct its diagonal elements $\widehat{V}_{\widehat{\gamma}}(q, q)$ for all q 's because we cannot define $\widehat{V}_{\widehat{\gamma}}(q, q)$ as (2).

To obtain substitutes for $\widehat{S}(q, q)$, we must borrow information across treatment arms. This motivates us to consider the following grouping strategy.

DEFINITION 4 (Grouping). Partition $\mathcal{Q}_U$ as $\mathcal{Q}_U = \bigcup_{g=1}^G \mathcal{Q}_{U,g}$ where $\mathcal{Q}_{U,g} \cap \mathcal{Q}_{U,g'} = \emptyset$ for all $g \neq g'$ and $|\mathcal{Q}_{U,g}| \geq 2$ for all $g \in [G]$. The partition should not depend on the observed data.

Definition 4 does not allow for data-dependent grouping, which can cause theoretical complications due to double-dipping into the data. Examples 8 and 9 below are special cases of Definition 4. By the construction in Definition 4, the $\mathcal{Q}_{U,g}$'s have no overlap, so we can also use $\langle g \rangle$ to denote $\mathcal{Q}_{U,g}$ and $\mathcal{G} = \{\langle g \rangle\}_{g=1}^G$ to denote the grouping strategy without causing confusions. Moreover, $|\langle g \rangle|$ must be larger than or equal to two so that there are at least two treatment levels in each $\langle g \rangle$. In general, we use $\langle g \rangle_q$ to indicate the group $\langle g \rangle$ that contains arm q , but when no confusion arises, we also simplify the notation to $\langle g \rangle$ if the corresponding q is clear from the context.

Define

$$\widehat{Y}_{\langle g \rangle} = \frac{1}{|\langle g \rangle|} \sum_{q \in \langle g \rangle} Y_q.$$

as the group-specific average, and construct

$$(23) \quad \widehat{V}_{\widehat{Y}}(q, q) = \mu_{\langle g \rangle} (Y_q - \widehat{Y}_{\langle g \rangle})^2 \quad \text{if } q \in \langle g \rangle$$

as the q th diagonal element of $\widehat{V}_{\widehat{Y}}$, where

$$\mu_{\langle g \rangle} = (1 - 2N^{-1})^{-1} (1 - |\langle g \rangle|^{-1})^{-2}$$

is a correction factor that is motivated by the theory below. Although the mean of $\widehat{Y}_{\langle g \rangle}$ has a simple formula, the mean of $\widehat{V}_{\widehat{Y}}(q, q)$ has a cumbersome form. We present a lower bound of $\mathbb{E}\{\widehat{V}_{\widehat{Y}}(q, q)\}$ below and relegate the complete formula to Appendix E.9 in the Supplementary Material. The results require Condition 5 below on the largest eigenvalue of the population correlation matrix of the potential outcomes in group $\langle g \rangle$, defined as

$$\varrho_{\langle g \rangle} = \varrho_{\max} \{ (S^*(q, q'))_{q, q' \in \langle g \rangle} \},$$

where S^* is the population correlation matrix of the potential outcomes.

CONDITION 5 (Bound on $\varrho_{\langle g \rangle}$). $N - \varrho_{\langle g \rangle} - (|\langle g \rangle| - 1) \geq 0$ for all $g \in \mathcal{G}$.

Condition 5 reflects a trade-off between N , $\langle g \rangle$ and $\varrho_{\langle g \rangle}$. It is more likely to hold with smaller correlations between arms within the same group and smaller subgroup sizes. By a natural bound $\varrho_{\langle g \rangle} \leq |\langle g \rangle|$, Condition 5 holds if $|\langle g \rangle| \leq (N + 1)/2$ for all $g \in [G]$. Examples 8 and 9 below satisfy Condition 5 automatically. With Condition 5, we can present Lemma 2 below.

LEMMA 2 (Sample mean and variance under grouping). *Assume grouping $\mathcal{G}$ according to Definition 4. We have*

$$\mathbb{E}\{\widehat{Y}_{\langle g \rangle}\} = \bar{Y}_{\langle g \rangle} \quad \text{where } \bar{Y}_{\langle g \rangle} = \frac{1}{|\langle g \rangle|N} \sum_{q \in \langle g \rangle} \sum_{i=1}^N Y_i(q) = \frac{1}{|\langle g \rangle|} \sum_{q \in \langle g \rangle} \bar{Y}(q).$$

Further assume Condition 5. We have

$$\mathbb{E}\{\widehat{V}_{\widehat{Y}}(q, q)\} \geq S(q, q) + \underbrace{\Omega(q, q)}_{\text{term III}} + \underbrace{\mu_{\langle g \rangle} (\bar{Y}(q) - \bar{Y}_{\langle g \rangle})^2}_{\text{term IV}},$$

where

$$(24) \quad \Omega(q, q) = \mu_{\langle g \rangle} |g|^{-2} \left(1 - \frac{\varrho_{\langle g \rangle}}{N} - \frac{|g| - 1}{N} \right) \sum_{q' \in \langle g \rangle, q' \neq q} S(q', q') \geq 0.$$

By Lemma 2, $\widehat{V}_{\widehat{Y}}(q, q)$, as an estimator for $S(q, q)$, is conservative, and the conservativeness depends on the variation of other arms q' that belong to $\langle g \rangle_q$ (term III) and the between-arm heterogeneity in means within $\langle g \rangle_q$ (term IV). We comment on some special cases below.

- If we assume homogeneity in means within subgroups, that is,

$$(25) \quad \bar{Y}(q) = \bar{Y}_{\langle g \rangle} \quad \text{for all } q \in \langle g \rangle,$$

then term IV in Lemma 2 vanishes. [Review Copy Only]

- If we assume homoskedasticity across treatment arms within the same subgroup, that is,

$$(26) \quad S(q, q) = S(q', q') \quad \text{for all } q, q' \in \langle g \rangle,$$

then term III in Lemma 2 becomes

$$(27) \quad \Omega(q, q) = \mu_{\langle g \rangle}(|g| - 1)|g|^{-2} \left(1 - \frac{\varrho_{\langle g \rangle}}{N} - \frac{|g| - 1}{N} \right) S(q, q).$$

Then we can combine (27) with $S(q, q)$ and use a smaller correction factor

$$\mu'_{\langle g \rangle} = (1 - |g|^{-1})^{-1} \{ (1 - |g|^{-1})(1 - 2N^{-1}) + |g|^{-1}(1 - (2|g| - 1)/N) \}^{-1} \leq \mu_{\langle g \rangle}$$

to reduce the conservativeness of variance estimation.

- If we assume the strong null hypothesis within subgroups, that is,

$$Y_i(q) = Y_i(q') \quad \text{for all } i \in [N] \text{ and } q, q' \in \langle g \rangle,$$

then both (25) and (26) hold. Applying the correction factor $\mu'_{\langle g \rangle}$, we can show $\widehat{V}_{\widehat{Y}}(q, q)$ is unbiased for $S(q, q)$: $\mathbb{E}\{\widehat{V}_{\widehat{Y}}(q, q)\} = S(q, q)$.

Lemma 2 suggests that ideally, we should group treatment arms based on the prior knowledge of the means and variances of the potential outcomes. While more general grouping strategies are possible, we give two examples for their simplicity of implementation. Both target the factorial design in Example 1.

EXAMPLE 8 (Pairing by the lexicographic order). Recall Example 1. We order the observations based on the lexicographical order of their treatment levels, then group the $(2k - 1)$ th level with the $(2k)$ th level ($1 \leq k \leq 2^{K-1}$). When $K = 3$, the grouping reduces to

$$\begin{aligned} \langle 1 \rangle &= \{(000), (001)\}, & \langle 2 \rangle &= \{(010), (011)\}, \\ \langle 3 \rangle &= \{(100), (101)\}, & \langle 4 \rangle &= \{(110), (111)\}. \end{aligned}$$

If the last factor has a small effect on the outcome, then we expect small differences in the mean potential outcomes within groups. As a sanity check, Condition 5 holds under this grouping strategy.

EXAMPLE 9 (Grouping based on a subset of the factors). Recall Example 1 again. If we have the prior knowledge that $K_0 < K$ factors are the most important ones, we can group the treatment levels based on these factors. Without loss of generality, assume that the first K_0 factors are the important ones. In particular, we can create $G = 2^{K_0} < Q$ groups, with each group $\langle g \rangle$ corresponding to treatment levels with the same important factors. Example 8 above is a special case with the first $K - 1$ factors as the important ones. Also, Condition 5 holds under this grouping strategy.

REMARK 1 (More on variance estimation). We have included two strategies for covariance estimation. On the one hand, they may seem ad hoc from a theoretical perspective. On the other hand, they are intuitive methods for covariance estimation. Covariance estimation without replications is a challenging problem in general. Therefore, the proposals can be viewed as first attempts for variance estimation in unreplicated designs. Indeed, more research efforts should be put into this problem. For instance, how do we compare different covariance estimation strategies? What is the “optimal” strategy for covariance estimation? We believe this is another independent project that goes beyond the scope of the current paper. In the current paper, we focus on the factorial design example and propose to apply two grouping strategies in Examples 8 and 9 that respect the structure of the design and are easy to implement. In particular, it is interesting to consider borrowing additional information such as pretreatment covariates to form better groups. Again, these directions require more technical work and go beyond the scope of the current paper.

Now we turn to the theoretical analysis of (23). Its properties depend on how successful the grouping $\mathcal{G}$ is, quantified by Condition 6 below.

CONDITION 6 (Bound on the within-group variation in potential outcome means). There exists a $\zeta > 0$, such that $\max_{g \in [G]} \max_{q \in \langle g \rangle} |\bar{Y}(q) - \bar{Y}_{\langle g \rangle}| \leq \zeta$.

The ζ in Condition 6 bounds the between-arm distance of the mean potential outcomes under grouping $\mathcal{G}$. It plays a key role in Theorem 7 below.

THEOREM 7 (Variance estimation for unreplicated designs). *Consider designs that satisfy Definition 3 with $|\mathcal{Q}_R| = |\mathcal{Q}_L| = 0$ and the covariance estimator in (23).*

(i) *Assume Condition 5. We have*

$$\mathbb{E}\{\hat{V}_{\hat{\gamma}}\} = V_{\hat{\gamma}} + \Omega + \text{Diag}\{\mu_{\langle g \rangle}(\bar{Y}(q) - \bar{Y}_{\langle g \rangle})^2\}_{q \in \mathcal{Q}_U} + N^{-1}(\Theta + S)$$

with $\Omega = \text{Diag}\{\Omega(q, q)\}_{q \in \mathcal{Q}_U}$ and $\Theta = \text{Diag}\{\Theta(q, q)\}_{q \in \mathcal{Q}_U}$, where the $\Omega(q, q)$'s are defined in (24) and the $\Theta(q, q)$'s are defined in (S66) in the Supplementary Material, satisfying $0 \leq \Theta(q, q) \leq 5\mu_{\langle g \rangle} \max_{q' \in \langle g \rangle} S(q', q')$. Therefore, $\mathbb{E}\{F^\top \hat{V}_{\hat{\gamma}} F\} \succeq V_{\hat{\gamma}}$.

(ii) *Assume Conditions 3 and 6. We have*

$$\|\hat{V}_{\hat{\gamma}} - \mathbb{E}\{\hat{V}_{\hat{\gamma}}\}\|_\infty^2 = O_{\mathbb{P}}\left\{\left(\max_{g \in [G]} \mu_{\langle g \rangle}\right)^2 \|F\|_\infty^4 N H^2\right\}.$$

(iii) *Assume Conditions 3 and 6. We have*

$$\|\hat{V}_{\hat{\gamma}} - \mathbb{E}\{\hat{V}_{\hat{\gamma}}\}\|_{\text{op}}^2 = O_{\mathbb{P}}\left\{\left(\max_{g \in [G]} \mu_{\langle g \rangle}\right)^2 \|F\|_\infty^4 N H^4\right\}.$$

Theorem 7(i) demonstrates that based on (23), the covariance estimator $\hat{V}_{\hat{\gamma}}$ is conservative for $V_{\hat{\gamma}}$, which implies that $F^\top \hat{V}_{\hat{\gamma}} F$ is conservative for the true covariance matrix of $\hat{\gamma}$. The conservativeness, however, has a more complex pattern compared with the setting with replications within all arms (Neyman (1923/1990), Imbens and Rubin (2015), Li and Ding (2017)). Theorem 7(i) shows three sources of conservativeness. The first part, captured by Ω , is due to the between-arm heteroskedasticity within each subgroup. However, it is fundamentally difficult to estimate each $S(q, q)$ without replications. The second part, captured by $\text{Diag}\{\mu_{\langle g \rangle}(\bar{Y}(q) - \bar{Y}_{\langle g \rangle})^2\}_{q \in \mathcal{Q}_U}$, is due to the between-arm heterogeneity in means within each subgroup. The part will be small if the grouping strategy ensures that the grouped arms have similar population averages of potential outcomes. The third part, captured by $N^{-1}(\Theta + S)$, is due to the difficulty of estimating S and in particular, the off-diagonal terms of S . The difficulty of estimating S has been well documented ever since Neyman (1923/1990) even in experiments with replications in each arm. It is possible to reduce this part but it requires additional assumptions, for example, the individual causal effects are constant.

Theorem 7(ii) and (iii) give the stochastic order of the estimation error of the covariance estimator $\hat{V}_{\hat{\gamma}}$ under the L_∞ norm and operator norm, respectively. If $\max_{g \in [G]} \mu_{\langle g \rangle}$, Δ and ζ are all constants, then

$$N \|\hat{V}_{\hat{\gamma}} - \mathbb{E}\{\hat{V}_{\hat{\gamma}}\}\|_\infty = O_{\mathbb{P}}(H/N^{1/2}), \quad N \|\hat{V}_{\hat{\gamma}} - \mathbb{E}\{\hat{V}_{\hat{\gamma}}\}\|_{\text{op}} = O_{\mathbb{P}}(H^2/N^{1/2}),$$

which gives sufficient conditions on H to ensure the convergence of $\hat{V}_{\hat{\gamma}}$ in L_∞ norm and operator norm, respectively.

Finally, equipped with the BEB on the quadratic form T in (19) and the conservative variance estimator studied in Theorem 7, it is immediate to establish Theorem 8 below for inference, which parallels Theorem 4.

THEOREM 8 (Wald-type inference under unreplicated design). *Consider the unreplicated design that satisfies $N_q = 1$ for all $q = 1, \dots, Q$ and Condition 1. Also assume (8) and Conditions 3–6. Let $N \rightarrow \infty$. If $H^{19/4}N^{-1/2} \rightarrow 0$, the Wald-type confidence set is asymptotically valid.*

3.3. General design. In this section, we consider general designs in Definition 3. First, we show a BEB on T in (19) in Theorem 9 below.

THEOREM 9 (Quadratic form BEB for general designs). *Consider the general design in Definition 3 that satisfies Condition 2 together with $\|F_L\|_\infty = O(Q^{-1})$ and $N = O(|Q_S|)$. Also assume (8). There exists a universal constant $C > 0$, such that*

$$(28) \quad \sup_{t \in \mathbb{R}} |\mathbb{P}(T \leq t) - \mathbb{P}(T_0 \leq t)| \leq C \frac{\max_{q \in [Q]} M_N(q)^3}{\{\min_{q \in Q_S} S(q, q)\}^{3/2}} \cdot \frac{H^{19/4}}{N^{1/2}}.$$

The proof of Theorem 9 relies on the general BEB Theorem S2. See Appendix E.13 for more details. Theorem 9 assumes $\|F_L\|_\infty$ and N has the same order as Q^{-1} and $|Q_S|$, respectively, which is helpful to establish the root- N convergence of BEB. We can relax the assumptions if we only need the CLT rather than the BEB. See Remark S3 in Appendix E.11 of the Supplementary Material. For ease of presentation, we omit the general results. A subtle feature of the upper bound in (28) is that $\max_{q \in [Q]} M_N(q)$ is the maximum value of the $M_N(q)$'s over all treatment arms whereas $\min_{q \in Q_S} S(q, q)$ is the minimum value of the $S(q, q)$'s over treatment arms in Q_S only.

Second, we construct a covariance estimator. It is a combination of the covariance estimators discussed in Sections 3.1 and 3.2. For the treatment arms with replications, we can calculate sample variances of the potential outcomes based on the observed data. For the treatment arms without replications Q_U , we need the grouping strategy in Definition 4. Therefore, we construct a diagonal covariance estimator $\widehat{V}_{\widehat{Y}}$ with the q th diagonal term

$$\widehat{V}_{\widehat{Y}}(q, q) = \begin{cases} \mu_{(g)}(Y_q - \widehat{Y}_{(g)})^2 & q \in Q_U, \\ \widehat{S}(q, q) & q \in Q_R \cup Q_L. \end{cases}$$

In a matrix form, it is equivalent to

$$\widehat{V}_{\widehat{Y}} = \begin{pmatrix} \widehat{V}_{\widehat{Y},U} & 0 & 0 \\ 0 & \widehat{V}_{\widehat{Y},R} & 0 \\ 0 & 0 & \widehat{V}_{\widehat{Y},L} \end{pmatrix},$$

where $\widehat{V}_{\widehat{Y},U}$, $\widehat{V}_{\widehat{Y},R}$, $\widehat{V}_{\widehat{Y},L}$ correspond to the diagonal covariance estimators for treatment arms Q_U , Q_R , Q_L , respectively. Recall the partitioning of F given in (14). Construct the final covariance estimator below:

$$(29) \quad \widehat{V}_{\widehat{Y}} = F^\top \widehat{V}_{\widehat{Y}} F = F_U^\top \widehat{V}_{\widehat{Y},U} F_U + F_R^\top \widehat{V}_{\widehat{Y},R} F_R + F_L^\top \widehat{V}_{\widehat{Y},L} F_L.$$

REMARK 2. The variance estimator (29) uses the second variance estimation strategy in Section 3.2.2 for the unreplicated design component. An extension of the first strategy in Section 3.2.1 is also feasible based on the following partition:

$$(30) \quad \begin{aligned} & F^\top \text{Diag}\{S(q, q)\}_{q \in [Q]} F \\ &= F_L^\top \text{Diag}\{S(q, q)\}_{q \in Q_L} F_L + F_R^\top \text{Diag}\{S(q, q)\}_{q \in Q_R} F_R \\ & \quad + F_U^\top \text{Diag}\{S(q, q)\}_{q \in Q_U} F_U \end{aligned}$$

The classical variance estimator for the first two terms of (30) is well-defined because each arm in $\mathcal{Q}_L$ and $\mathcal{Q}_R$ still contains at least two units. For the last part, the variance estimator can be constructed in a similar way to (22). We omit the details.

The decomposition in (29) allows us to characterize the statistical properties of $\widehat{V}_{\widehat{\mathcal{Y}}}$ by combining the results from Sections 3.1 and 3.2.

THEOREM 10 (Covariance estimation for general designs). *Consider the designs in Definition 3 and the covariance estimator in (29). Assume Conditions 3, 5 and 6. Assume $\max_{g \in [G]} \mu_{\langle g \rangle}$, Δ and ζ are constants.*

- (i) $\mathbb{E}\{\widehat{V}_{\widehat{\mathcal{Y}}}\} \geq V_{\widehat{\mathcal{Y}}}$.
- (ii) We have

$$\|\widehat{V}_{\widehat{\mathcal{Y}}} - \mathbb{E}\{\widehat{V}_{\widehat{\mathcal{Y}}}\}\|_{\infty}^2 = O_{\mathbb{P}}(\|F_U\|_{\infty}^4 |\mathcal{Q}_U| H^2 + \|F_R\|_{\infty}^4 |\mathcal{Q}_R| H^2 + \|F_L\|_{\infty}^4 |\mathcal{Q}_L|^4 N_L^{-3} H^2).$$

- (iii) We have

$$\|\widehat{V}_{\widehat{\mathcal{Y}}} - \mathbb{E}\{\widehat{V}_{\widehat{\mathcal{Y}}}\}\|_{\text{op}}^2 = O_{\mathbb{P}}(\|F_U\|_{\infty}^4 |\mathcal{Q}_U| H^4 + \|F_R\|_{\infty}^4 |\mathcal{Q}_R| H^4 + \|F_L\|_{\infty}^4 |\mathcal{Q}_L|^4 N_L^{-3} H^4).$$

In Theorem 10, we assume $\max_{g \in [G]} \mu_{\langle g \rangle}$, Δ and ζ to be constants to simplify the presentation. Without this assumption, we can derive results similar to those in Theorem 7 but relegate finer results to Appendix E.12 in the Supplementary Material. Theorem 10(i) shows the conservativeness of $\widehat{V}_{\widehat{\mathcal{Y}}}$ as a direct consequence of Theorems 3(i) and 7(i). Theorem 10(ii) and (iii) show the stochastic order of the estimation error of $\widehat{V}_{\widehat{\mathcal{Y}}}$ in L_{∞} norm and operator norm, respectively. We only discuss Theorem 10(ii) below. If $\|F\|_{\infty} = O(Q^{-1})$ as in the factorial design in Example 1, it reduces to

$$(31) \quad \|\widehat{V}_{\widehat{\mathcal{Y}}} - \mathbb{E}\{\widehat{V}_{\widehat{\mathcal{Y}}}\}\|_{\infty} = O_{\mathbb{P}}\{Q^{-2} H(|\mathcal{Q}_U|^{1/2} + |\mathcal{Q}_R|^{1/2} + |\mathcal{Q}_L|^2 N_L^{-3/2})\}.$$

Therefore, if N and Q are of the same order, then $N\|\widehat{V}_{\widehat{\mathcal{Y}}} - \mathbb{E}\{\widehat{V}_{\widehat{\mathcal{Y}}}\}\|_{\infty} = O_{\mathbb{P}}(HN^{-1/2})$. Besides, when one or two of $\mathcal{Q}_U$, $\mathcal{Q}_R$, $\mathcal{Q}_L$ are small or absent, the stochastic orders in Theorem 10 still hold because the large terms in (31) will dominate the rest. In particular, if $|\mathcal{Q}_U| = |\mathcal{Q}_R| = 0$, then $\|\widehat{V}_{\widehat{\mathcal{Y}}} - \mathbb{E}\{\widehat{V}_{\widehat{\mathcal{Y}}}\}\|_{\infty} = O_{\mathbb{P}}(HN_L^{-3/2})$, which gives the same rate as Theorem 3. If $|\mathcal{Q}_L| = 0$, then we should interpret $0 \cdot \infty = 0$ in Theorem 10(ii) to obtain $\|\widehat{V}_{\widehat{\mathcal{Y}}} - \mathbb{E}\{\widehat{V}_{\widehat{\mathcal{Y}}}\}\|_{\infty} = O_{\mathbb{P}}(HQ^{-3/2}) = O_{\mathbb{P}}(HN_S^{-3/2})$, which also agrees with Theorem 3.

Finally, the BEB on the quadratic form T and the conservativeness of the covariance estimator ensure Theorem 11 below for inference.

THEOREM 11 (Wald-type inference under general designs). *Consider the general design in Definition 3 that satisfies Condition 2 together with $\|F_L\|_{\infty} = O(Q^{-1})$ and $N = O(|\mathcal{Q}_S|)$. Also assume (8) and Conditions 3–6. Let $N \rightarrow \infty$. If $H^{19/4} N^{-1/2} \rightarrow 0$, the Wald-type confidence set is asymptotically valid.*

This concludes our discussion of design-based causal inference with possibly a diverging number of treatment levels and varying group sizes across treatment levels.

4. Discussion. We provide general BEBs for design-based causal inference that can accommodate possibly diverging treatment levels and varying group sizes. They serve as the theoretical foundation for causal inference in complex experiments. When we were polishing the paper, Shi, Wang and Ding (2025) and Masoero et al. (2024) used our BEBs to analyze factorial designs and multiple randomization designs, respectively. We look forward to see more applications of our theoretical results in future research.

Our paper is mainly theoretical. Nevertheless, we use simulation studies to evaluate finite-sample properties of the estimators as well as their variance estimators. We relegate the details to Appendix A in the Supplementary Material. It is of interest to see more comprehensive simulation studies and concrete empirical applications in the future.

We focused on scalar outcomes. Results for vector outcomes are also important in both theory and practice. Li and Ding (2017) reviewed CLTs and many applications with vector outcomes under the regime of a fixed number of treatment levels and large sample sizes within all treatment levels. We include an extension of the BEB for vector outcomes under a general regime; see Appendix D.5 in the Supplementary Material.

Asymptotic results for design-based inference are often criticized because the population of interest is finite but the asymptotic theory requires a growing sample size. Establishing BEBs is an important theoretical step to characterize the finite-sample performance of the statistics. Alternatively, it is also desirable to derive nonasymptotic concentration inequalities for the estimators under the randomization model. This requires a deeper understanding of sampling without replacement and permutation statistics. We leave it to future research.

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SUPPLEMENTARY MATERIAL

Supplement to “Berry–Esseen bounds for design-based causal inference with possibly diverging treatment levels and varying group sizes” (DOI: [10.1214/25-AOS2569SUPP](https://doi.org/10.1214/25-AOS2569SUPP.pdf); .pdf). The Supplementary Material contains additional results on simulation results, general linear permutation statistics, randomization-based inference and all the technical proofs.

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